



AgroVoice-XAI: Smart Visual Crop Advisory

PHASE II FINAL TECHNICAL SUMMARY

Multi-Modal Deep Learning, Explainable AI (Grad-CAM), Microclimate Telemetry & Indic Vernacular Clinical Prescriptions

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98.43%

VALIDATION ACCURACY



22 ms

INFERENCE LATENCY



39 Taxa

DIAGNOSED CLASSES



54,305

CURATED LEAF IMAGES



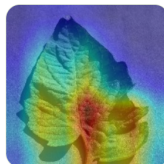
1. VISUAL EXPLAINABLE AI (GRAD-CAM)

PyTorch Hook

Neural Saliency Grounding: Intercepts forward activations & backward gradients at features[-1] to localize active fungal lesions with zero human bias.



Your Photo



Infection Spots (Grad-CAM)

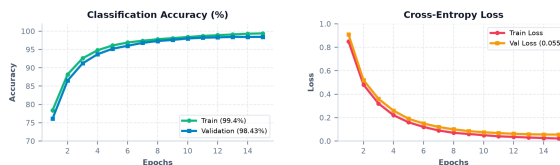
How the AI Sees the Leaf (Plain Language):

- Red & Yellow Hotspots:** Highlights the exact active fungal infection spots and necrosis.
- Cool Blue Regions:** Indicates healthy green leaf areas that the AI safely ignored.
- Visual Proof (XAI):** Gives farmers 100% confidence that the diagnosis is real.



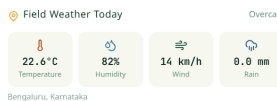
2. MODEL CONVERGENCE & LOSS GRAPHS

MobileNetV2



3. WEATHER RISK & DISEASE TRIANGLE

Open-Meteo API



Telemetry	Value	Pathogen Risk Impact
Humidity	82% RH	Accelerates Spore Germination
Temperature	22.6°C	Optimal Fungal Incubation
Wind Speed	14 km/h	Moderate Spore Dispersion



4. CLINICAL AGRONOMIC ENGINE

KVK Verified

- Etiology / Cause:** Fungal spore infection entering via moisture.
- Organic:** Gomutra, Neem Oil 5ml/L, Buttermilk & Trichoderma.
- Chemical Spray:** Saaf / Ridomil @ 30g per 15L backpack pump.
- Cultural Prevention:** Crop rotation, soil aeration, plucking burnt leaves.



5. MULTILINGUAL VOICE & DELIVERY

Web Speech API

ಕನ್ನಡ: "15 ಲೀಟರ್ ತೆಂಕಿ 30 ಗ್ರಾಂ ಸಾಫ್ ಬೆರೆಸಿ ಸಿಂಪಡಿಸಿ."
हिन्दी: "15 लीटर टंकी में 30 ग्राम साफ मिलाकर छिड़कें।"
English: "Recommended 15L pump dosage: 30g Saaf."

- Official PDF Health Card:** 1-Click print prescription with GPS & dosages.
- 1-Tap WhatsApp Share:** Sends diagnosis straight to local pesticide shop.